

Hypothesis Test Notes

Objective

The objective of this experiment is to determine whether the new onboarding campaign (Treatment group) significantly improves the Paid Conversion Rate compared to the existing onboarding experience (Control group).

Primary Metric Tested

Paid Conversion Rate

Paid Conversion Rate is calculated as:

$$\text{Paid Conversion Rate} = \text{Number of Paid Users} \div \text{Total Users}$$

This metric was selected because it directly measures the business objective of converting free users into paying customers, making it the North Star Metric for the experiment.

Business Question

Should the company launch the new onboarding campaign to all users?

The answer depends on whether the Treatment group demonstrates a statistically significant improvement in Paid Conversion Rate without negatively affecting important guardrail metrics.

Hypotheses

Null Hypothesis (H_0)

There is **no statistically significant difference** in Paid Conversion Rate between the Control and Treatment groups.

Mathematically:

$$H_0: p_{\text{Treatment}} = p_{\text{Control}}$$

Alternative Hypothesis (H_1)

The Treatment group has a **higher Paid Conversion Rate** than the Control group.

Mathematically:

$H_1: p_{\text{Treatment}} > p_{\text{Control}}$

Test Type

One-tailed Two-Proportion Z-Test

A one-tailed test is appropriate because the business objective is specifically to determine whether the Treatment improves conversion compared to the Control.

Significance Level

$\alpha = 0.05$

Decision rule:

- If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$
 - If $p\text{-value} \geq 0.05 \rightarrow \text{Fail to Reject } H_0$
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Data Used

The hypothesis test compares:

- Number of users in Control
- Number of users in Treatment
- Number of paid conversions in each group

These values are obtained from the experiment dataset after data cleaning and validation.

Assumptions

The hypothesis test assumes:

- Users were randomly assigned to Control and Treatment groups.
 - User observations are independent.
 - Sample size is sufficiently large.
 - Paid Conversion is a binary outcome (0 = No, 1 = Yes).
 - No significant data quality issues remain after cleaning.
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Test Procedure

1. Clean and validate the dataset.
 2. Calculate Paid Conversion Rate for each group.
 3. Perform a Two-Proportion Z-Test.
 4. Calculate the p-value.
 5. Compare the p-value with $\alpha = 0.05$.
 6. Make the statistical decision.
 7. Interpret the result in a business context.
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Decision Rule

If:

$p\text{-value} < 0.05$

Reject the Null Hypothesis.

This indicates that the Treatment group performs significantly better than the Control group.

Otherwise:

Fail to Reject the Null Hypothesis.

There is insufficient evidence to conclude that the Treatment improves Paid Conversion Rate.

Business Interpretation

If the hypothesis test is statistically significant, the company has evidence that the new onboarding campaign increases Paid Conversion Rate.

However, the launch decision should not rely solely on conversion improvement.

The following guardrail metrics must also be evaluated:

- Refund Rate
- Support Ticket Rate
- Average Engagement Score
- Average Days to Convert
- Revenue Quality
- Segment-Level Performance

If these guardrail metrics remain stable or improve, the Treatment can be recommended for rollout.

Guardrail Evaluation

The following metrics are reviewed before making the final recommendation:

Guardrail Metric	Purpose
Refund Rate	Ensure higher conversions are not creating dissatisfied customers
Support Ticket Rate	Measure customer friction
Average Engagement Score	Confirm users remain engaged
Average Days to Convert	Evaluate onboarding efficiency
Revenue Quality	Ensure sustainable revenue growth
Segment-Level Analysis	Identify any customer groups negatively affected

Final Interpretation

The experiment should be considered successful only if:

- Paid Conversion Rate increases significantly.
- Statistical significance is achieved (p-value < 0.05).
- No guardrail metric deteriorates materially.
- No important customer segment performs significantly worse than the Control group.

If all conditions are satisfied, the recommended business action is:

Launch the new onboarding experience for all users.

Otherwise, the recommendation should be either:

- Continue testing
- Perform a phased rollout
- Launch only for selected user segments

depending on the observed results.